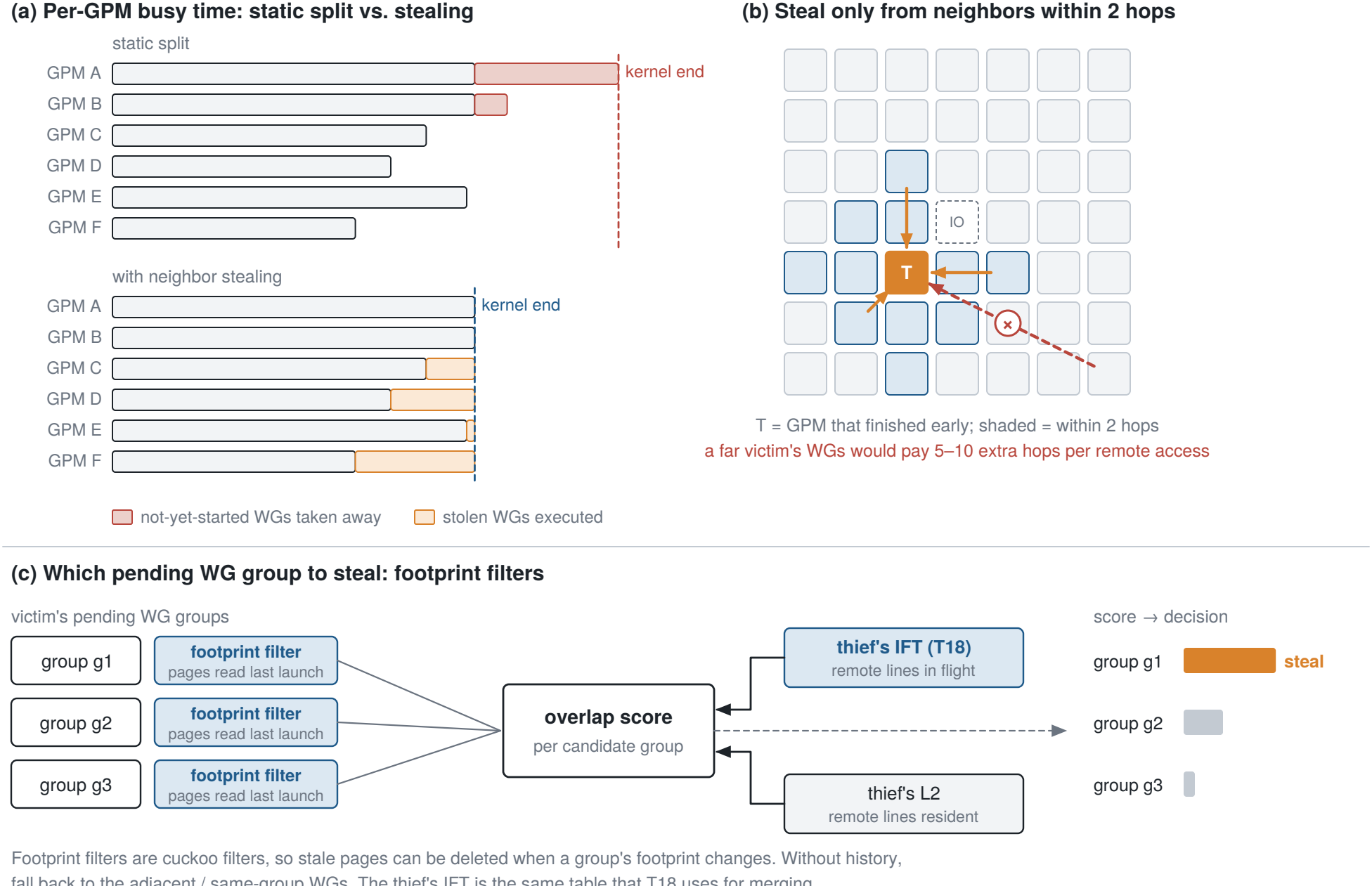
(a) Per-GPM busy time: static split vs. stealing
static split
GPM A
GPM B
GPM C
GPM D
GPM E
GPM F
kernel end
with neighbor stealing
GPM A
GPM B
GPM C
GPM D
GPM E
GPM F
kernel end
not-yet-started WGs taken away
stolen WGs executed
(b) Steal only from neighbors within 2 hops
IO
T
T = GPM that finished early; shaded = within 2 hops
a far victim's WGs would pay 5–10 extra hops per remote access
(c) Which pending WG group to steal: footprint filters
victim's pending WG groups
group g1
group g2
group g3
footprint filter
pages read last launch
footprint filter
pages read last launch
footprint filter
pages read last launch
overlap score
per candidate group
thief's IFT (T18)
remote lines in flight
thief's L2
remote lines resident
score → decision
group g1
steal
group g2
group g3
Footprint filters are cuckoo filters, so stale pages can be deleted when a group's footprint changes. Without history,
fall back to the adjacent / same-group WGs. The thief's IFT is the same table that T18 uses for merging.